

# An Agentic AI Approach with humans in the loop: Exploring the case of climate education through evidence synthesis - Appendices

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# 1 Appendix A: Examples of Climate Literacy Impact Anchor statements

```
1 clii_anchors = {
2   "Cognitive": [
3     "knowledge about climate change, e.g. Knowing that the
4       melting of permafrost affects the Earth's climate or
5       understanding the role of roof gardens in reducing
6       carbon dioxide levels, notes UNCC:Learn",
7     "understanding of environmental issues e.g. Recognizing
8       that burning fossil fuels for power, cutting down
9       forests, and manufacturing goods are major causes of
10      climate change and that these actions lead to effects
11      like rising temperatures and extreme weather events,
12      according to the United Nations and the National
13      Oceanic and Atmospheric Administration (NOAA).",
14     "climate literacy and awareness, e.g. analyzing news
15      articles or social media posts about climate change,
16      discerning scientific consensus from misinformation,
17      and understanding the uncertainties in scientific
18      findings"
19   ],
20   "Affective": [
21     "emotional response to climate change, e.g. This category
22       includes a range of emotions such as eco-anxiety, a
23       chronic fear of environmental doom; guilt about
24       personal lifestyle choices; sadness and grief over the
25       loss of ecosystems; and anger at inaction from
26       governments and corporations. These negative emotions
27       can be contrasted with hope and optimism, which can
28       also motivate positive action.",
29     "motivation to take action, e.g. The motivation to act on
30       climate change is driven by a sense of duty and
31       responsibility to future generations. It also relies on
32       a person's efficacy beliefs, or the conviction that
33       their actions can make a difference. Health and safety
34       concerns, when climate change is framed as a direct
35       threat to personal well-being, are also powerful
36       motivators.",
37     "concern and care for the environment, e.g. A person's
38       level of concern is strongly linked to their emotional
39       connection to nature and a deep sense of belonging to
40       the natural world. Conversely, psychological distancing
41       ---the feeling that climate change is a problem for
42       someone else---can reduce a person's concern. Finally,
43       social norms and influence play a role, as people are
44       often motivated to adopt pro-environmental behaviors to
45       align with their social group. "
46   ],
47   "Behavioral": [
```

13 "Actions to mitigate climate change: These encompass both  
 individual and large-scale actions aimed at reducing or  
 preventing greenhouse gas emissions. Examples include  
 reducing personal energy consumption, adopting  
 renewable energy sources, and switching to sustainable  
 transport like walking, biking, or electric vehicles.  
 It also involves lifestyle changes such as eating more  
 plant-based foods, reducing food waste, and consciously  
 minimizing consumption of goods like fast fashion and  
 plastics.",

14 "Community participation in sustainability: This involves  
 the active and genuine involvement of people in  
 addressing local problems and implementing projects to  
 promote sustainable development. Participation can be  
 driven by a motivation to improve one's community,  
 building a sense of collective purpose. Strategies  
 include community needs assessments, mobilization, and  
 working with local authorities. It is crucial that  
 participation is inclusive, involving women and other  
 vulnerable groups, and that agencies or governments  
 respect local knowledge and needs.",

15 "Adoption of eco-friendly practices: The adoption of eco-  
 friendly practices by individuals and organizations  
 includes a range of actions that minimize environmental  
 harm, reduce waste, and conserve energy. Examples  
 involve using reusable items, recycling, upcycling,  
 composting, and conserving water. In the corporate  
 sector, this includes implementing strategies to reduce  
 a company's carbon footprint, investing in sustainable  
 supply chains, and offering eco-friendly products."

16 ],

17 "Institutional": [  
 18 "policy development for climate adaptation: Effective  
 policy-making for climate change adaptation involves  
 national and sub-national governments developing  
 strategies and measures to reduce vulnerability to  
 climate impacts. This includes implementing large-scale  
 measures like enhancing early warning systems,  
 strengthening infrastructure, and developing strategies  
 for managing water, land, and food systems. It also  
 requires a 'whole-of-government' approach and  
 integrating climate considerations into development  
 plans.",

19 "Governance and institutional support: This involves  
 building robust institutional arrangements and  
 governance models to address social problems like  
 climate change and ensure sustainable development. Good  
 governance is characterized by participatory behaviour  
 , transparency, accountability, and coordination across  
 different levels of government. Institutional support,  
 often technical and strategic, is aimed at

strengthening government departments to improve financial management, service delivery, and overall effectiveness.",

"Capacity building in organizations: This is the process of helping individuals, organizations, and communities strengthen their skills, knowledge, and systems to achieve their goals. For organizations, this involves upskilling and reskilling employees, developing leadership, forming partnerships, and upgrading technology and processes. Ultimately, capacity building aims to transform behaviours and improve an organization's ability to achieve its mission."

]

}

## 2 Appendix B: A list of queries used for querying data

```
1
2 queries = """
3         tree planting initiatives South Africa,
4         coastal conservation South Africa,
5         wetland restoration South Africa,
6
7         climate change education South Africa,
8         environmental curriculum development South Africa,
9         climate change workshops South Africa,
10        sustainability training South Africa,
11        climate change awareness campaigns South Africa,
12        community climate resilience South Africa,
13        environmental activism South Africa
14        """
```

### 3 Appendix C: Data extraction prompts

```
1 data_extraction_prompt_for_interventions = """
2 You are an AI assistant specializing in identifying and
3   classifying content related to **Sustainable Development Goal
4   13 (Climate Action)** --- specifically targets 13.3.
5 Your task is to follow the instructions as listed below:
6
7 If the text **does not reference any climate change-related
8   concepts or ideas **, mark it as **NOT RELATED TO CLIMATE
9   CHANGE**.
10
11 ---
12
13 Document:
14 <document>
15 {document}
16 </document>
17
18 Chunk to analyze:
19 <chunk>
20 {chunk}
21 </chunk>
22
23 ---
24
25 Instructions:
26 1. Read the entire document carefully.
27 2. Produce a detailed summary organised under the following
28   headings:
29   - Context & Setting: Where the work takes place (region e.g. (
30     Mpumalanga, uMkhanyakude District Municipality),
31     communities, ecosystems, timeframe).
32   - Specific country: The country where this work has taken
33     place (e.g., South Africa, India, USA).
34   - Population / Stakeholders: Who is affected or involved (e.g
35     ., local communities, schools, government agencies).
36   - Interventions or Actions: Specific climate-related
37     interventions such as:
38     Mitigation (e.g., reducing greenhouse gas emissions),
39     Adaptation (e.g., preparing for climate impacts),
40     Ecosystem-based solutions (tree planting, wetland
41     restoration, coastal conservation),
42     or Awareness & Education Efforts.
43   - Outcomes & Impacts: Quantitative or qualitative impacts---
44     include numbers if mentioned (e.g., number of trees planted
45     , training sessions held, policy changes).
46   - Links to SDG Targets: Explicit or implied alignment with SDG
47     13.3.
48 3. Write the summary as neutral factual prose, NOT bullet points.
```

```

36 4. For dates, Paraphrase for clarity but do not invent dates or
    targets.
37 5. If no specific date is mentioned, indicate it as "Not specified
    "
38
39 {{
40   "summary": "<Use the detailed summary to populate this section
        or if not related to climate change output 'NOT RELATED TO
        CLIMATE CHANGE'>",
41   "community": "<Who or what is affected? e.g., South African
        communities, ecosystems, schools>",
42   "project_name": "<What is the name of the project? e.g. Green
        Roots Initiative, Trees for Tomorrow, Urban Canopy Project,
        Blue Coast Guardians, Wetlands for Wildlife>",
43   "time_frame": "<When the project started, when it is planned to
        end, or any deadlines mentioned.>",
44   "future_goals": "<Any objectives, targets, or milestones the
        project aims to achieve in the future.>",
45   "intervention": "<What action / intervention is described? e.g.,
        tree planting, workshops, curriculum development>",
46   "outcome": "<What is achieved or changed? e.g., increased
        awareness, restored wetlands, reduced emissions.>",
47   "impact": "<Impacts explicitly mentioned in the text --- can be
        numerical (e.g., '5000 trees planted') or qualitative (e.g.,
        'improved community knowledge.')>",
48   "sdg_13_3": "<If the intervention aligns with SDG 13.3, answer
        YES or NO and add a reason.>",
49   "sa_project": "<If this is a South African initiative output YES
        otherwise NO.>",
50   "climate_related": "<If this is a climate change related, output
        YES otherwise NO.>"
51 }}
52
53 Only return the JSON object --- no explanation, no extra
    commentary.
54
55 ""

```

## 4 Appendix D: Code and prompts for generating semantic similarity and sentiment analysis heat map

```
1 import json
2 from tqdm import tqdm
3
4 def llm_judge_sentiment_temporal(text, field_name):
5     if not text or text.strip().lower() in ["not specified", "none", ""]:
6         return {"sentiment": "neutral", "timeframe": "present"}
7
8     prompt = f"""
9     You are analyzing text extracted from a climate-related
10     research article.
11     The text is from the **{field_name}** section.
12
13     Determine:
14     1. Sentiment polarity: "positive", "negative", or "neutral"
15     2. Temporal context: "past", "present", or "future"
16
17     Respond strictly in JSON format, for example:
18     {{
19         "sentiment": "positive",
20         "timeframe": "future"
21     }}
22
23     Text: "{text}"
24     """
25
26     try:
27         messages = [{"role": "user", "content": prompt}]
28         response = client.invoke(messages)
29         return response.content.strip()
30     except Exception as e:
31         print(f"Error for {field_name}: {e}")
32         return {"sentiment": "neutral", "timeframe": "present"}
33
34 def safe_parse_response(response):
35     """Ensure the LLM result is a dict with 'sentiment' and 'timeframe'."""
36     if isinstance(response, dict):
37         return response
38     if isinstance(response, str):
39         try:
40             # try parsing JSON string
41             return json.loads(response)
42         except Exception:
43             pass
44     # fallback in case of error
45     return {"sentiment": "neutral", "timeframe": "unspecified"}
```



```

46 fields = ["interventions", "outcomes", "impacts", "time_frames", "
    future_goals"]
47 error_data = []
48 for field in tqdm(fields):
49     try:
50
51         print(f"Processing {field}...")
52         results = new_data_df[field].apply(
53             lambda x: safe_parse_response(llm_judge_sentiment_temporal
54                 (str(x), field))
55         )
56
57         new_data_df[f"{field}_sentiment"] = results.apply(lambda r: r.
58             get("sentiment", "neutral"))
59         new_data_df[f"{field}_timeframe"] = results.apply(lambda r: r.
60             get("timeframe", "unspecified"))
61
62     except Exception as e:
63         error_data.append({"field": field, "error": str(e), "results":
64             results})
65         print(f"Error for {field}: {e}, results: {results}")
66         continue

```